

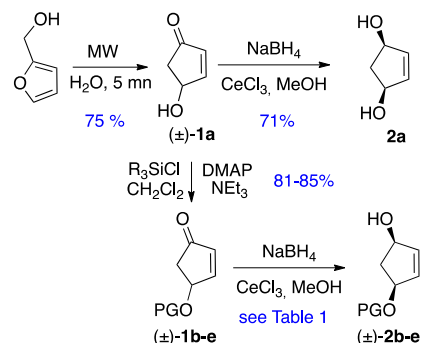
Scheme 2. Chemoenzymatic route to cyclopentanoid derivatives from furfuryl alcohol via 4-silyloxycyclopent-2-en-1-one enantiomers.

the chirality being introduced through enzymatic desymmetrization of cyclopent-4-en-1,3-diyl diacetate with electric eel cholinesterase.^{9j} Despite various publications dealing with asymmetric access to **1a** and some of its derivatives, there is no systematic comparison and no data showing that sequences affording chiral 4-*tert*-butyldimethylsilyloxycyclopent-2-en-1-one are still efficient for their TIPS analogs. We thus reinvestigated some of these routes and reported here our own findings, starting from the cheap 2-hydroxymethylfuran, conveniently leading to each enantiomer in high yields and ee's (Scheme 2).

As reported,¹⁶ 2-hydroxymethylfuran can be hydrated in warm acidic phosphate buffer. However, the useful hydroxycyclopentenone (±)-**1a** can only be isolated with modest yields (~40%) since numerous side-products are also formed due to the acid-sensitivity of both the starting alcohol and the product. Furthermore, (±)-**1a** is water-soluble and difficult to extract from the resulting mixture. Those problems could be circumvented by performing the reaction in pure water at 0.2 M concentration without any additive under microwave irradiation for 5 min,¹⁷ directly followed *without any work-up* by water evaporation and a rapid chromatography. Hydroxycyclopentenone (±)-**1a** can thus be isolated with drastically improved efficiency in 75% yield on multigram scale (Scheme 3).

Although the racemic hydroxycyclopentenone (±)-**1a** and its esters can directly be resolved,⁷ higher yields and ee's are often achieved by desymmetrization of the corresponding *meso* diol *cis*-**2a** and its esters.⁹ However, highly stereoselective reduction of **1a** is required to get the *cis* and not the *trans* isomer, the former being required for desymmetrization due to its *meso* symmetry. Although already studied,^{8,18} we found that this reduction could be performed with high selectivity on the free hydroxycyclopentenone (±)-**1a** and on its protected analogs (±)-**1b–e**. In the latter cases, the selectivity proved dependent on the size of the protecting group at the 4-position (Table 1).

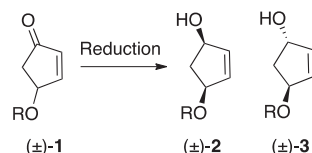
Reported with good yield and *cis*-*trans* selectivity (entry 1),⁸ the reduction of the free (±)-**1a** could largely be improved under adjusted Luche conditions with excess of cerium chloride at low temperature. Due to the high water solubility of the diol, aqueous work-up was avoided by only adding silica to destroy or trap reagents and products. Solvent evaporation and solid deposit chromatography then provided the diols in almost quantitative yields and high selectivity, allowing to isolate pure *cis*-**2a** in 71%¹⁹ yield (entry 2). With the trimethylsilyl group, deprotection occurred under Luche conditions, but the use of DIBAH in toluene proved effective enough in terms of yield, although not as selective (entry 3 vs 2). Both methods (and others) have been reported for the *tert*-butyldimethylsilyl (TBDMS) analog (±)-**1c** and again the Luche conditions were more efficient (entry 5 vs 4) but the selectivity was lower than the one achieved without the protecting group (entry 5 vs 2). With the *tert*-butyldiphenylsilyl (TBDPS) analog (±)-**1d**, similar results and conclusion were obtained (entries 6–7), as well as with the very bulky triisopropylsilyl (TIPS) analog (±)-**1e** (entries



Scheme 3. Route to 4-hydroxycyclopent-2-en-1-one and its derivatives.

Table 1

Screening of conditions for the reduction of 4-hydroxycyclopent-2-en-1-one, protected or not



Entry	R	Conditions	Yield ^a (%) <i>cis:trans</i> ^b
1 ^c	H	NaBH ₄ 1 equiv; CeCl ₃ 1 equiv MeOH-THF, rt	68 7:1
2	1a	NaBH ₄ 1 equiv; CeCl ₃ 2 equiv MeOH, –20 °C	95(71) 10:1
3	SiMe ₃ 1b	DIBAH 1.1 equiv Tol., –20 °C	78 ^d 5:1
4 ^c	Si ^t BuMe ₂ 1c	DIBAH 1.1 equiv Tol., –40 °C	72 9:1
5 ^c	1c	NaBH ₄ , 0.5 equiv; CeCl ₃ , 2 equiv MeOH, –20 °C	93 8:1
6	Si ^t BuPh ₂ 1d	DIBAH 1.1 equiv Tol., –40 to –20 °C	81 3.5:1
7	1d	NaBH ₄ , 1.1 equiv; CeCl ₃ , 2 equiv MeOH, rt	84 6.4:1
8	Si ⁱ Pr ₃ 1e	DIBAH 1.1 equiv Tol., –40 °C	88 5.8:1
9	1e	NaBH ₄ , 1.1 equiv; CeCl ₃ , 2 equiv MeOH, –20 °C	82(74) 8:1

^a Isolated yields (yield of pure *cis* isomer).

^b Determined by ¹H NMR on the crude mixture.

^c From Ref. 8

^d Isolated as deprotected alcohol.

8–9). Nevertheless, the good yield and selectivity achieved with the TIPS group allowed us to isolate the pure (±)-*cis*-**2e** in 74% yield (entry 9).²⁰

The *meso* diol *cis*-**2a** was then submitted to acylating agents (vinyl acetate (VA) or vinyl butyrate (VB)) in the presence of various enzymes, leading to the corresponding mono- or diesters (Scheme 4). The latter were also selectively hydrolyzed by enzymes. Interestingly, each route provided each enantiomer.

The enantioselective acylation of *meso* diol *cis*-**2a** has been reported with pancreatin as biocatalyst.^{9a} However, the reported conditions led to the acetate (–)-**2f** with a good enantioselectivity but not high enough for synthetic purposes (Table 2, entry 1). Using another source of enzyme provided (–)-**2f** in very high ee in a reasonable time, but to the detriment of yield due to the increased formation of the *meso* diacetate **4f** (entry 2).²¹ Increasing the reaction time under the same conditions or using vinyl butyrate allowed to get the enantiopure monoesters **2f–g**, but still with the same modest yields (entries 3 and 4).